

Eco-friendly method for catalytic aerial oxidation of Aloe-emodin to rheinal: An intermediate for diacerein¹

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An eco-friendly process for the synthesis of 4,5-dihydro-9,10-dioxo-9,10-dihydroanthracene-2-carbaldehyde (rheinal) from 1,8-dihydroxy-3-(hydroxymethyl)-9,10-anthracenedione (aloe-emodin) *via* aerial oxidation in presence of catalysts such as oxovanadium, copper(II) salen or enzymes containing copper(II) *i.e.* laccase is described.

Keywords: Aloe-emodin, rheinal, oxovanadium, Laccase, copper (II) salen H₄

Osteoarthritis is the most common joint disorder in the world; the most common form of arthritis². It is one of the most frequent causes of pain, loss of function and disability in adults, second only to Ischemic Heart Disease (IHD)³. Osteoarthritis is a degenerative joint disease characterized by fibrillation, thinning and erosion of articular cartilage, depletion of proteoglycan, abnormal replication of chondrocytes and formation of osteophytes at joint margins⁴.

Diacerein or diacetylrhein (4,5-diacetoxy-9,10-dihydro-9,10-di-oxo-2-anthracene carboxylic acid) is a newly introduced symptomatic slow acting drug (or, chondroprotective drug) for treatment of Osteoarthritis⁵. It is a semi-synthetic anthraquinone derivative of glycoside Aloin, also known as Barbaloin, obtained from *Aloe* species.

The active metabolite of diacerein is rhein (4,5-dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carboxylic acid), which inhibits the production of IL-1 β by human monocytes *in vitro*. IL-1 mediated enhancement of collagenase production by joint chondrocytes is also reduced by diacerein. It also decreases the number of urokinase receptors on chondrocytes to near normal levels and reduces fibrinolytic activity in synovial fluid. Leukocyte migration, lysosomal enzyme release, super oxide production and chemotaxis are also inhibited by rhein in dose dependent manner²⁻⁵.

Industrial process for preparation of diacerein comprises acetylation of aloin, which is a naturally occurring glycoside and subsequent chromic

oxidation of the acetylated product to obtain diacerein⁶. In the literature, various methods are reported for the purification of crude diacerein obtained by chromic oxidation⁷⁻¹⁰. However, these processes suffer from several drawbacks such as complex multi-step processes and use of toxic solvents/chemicals, which result in decrease in the yield of pure diacerein. Moreover, hexavalent chromium compounds such as chromic anhydride and dichromates are known for their high toxicity and carcinogenicity as well as harmful effects on the environment¹¹.

Chromium free oxidation of aloe-emodin (aglycone part of aloin) by using sodium nitrite and large volume of sulfuric acid¹²; and TEMPO catalyzed oxidation of benzyl protected aloe-emodin have been reported¹³.

Another approach for synthesis of diacerein was through 4,5-dihydroxy-9,10-dioxo-9,10-dihydroanthracene-2-carbaldehyde (rheinal) which was converted into 4,5-dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carboxylic acid (rhein), key intermediate for diacerein. Reported processes for preparation of rheinal are by oxidation of aloe-emodin through Swern oxidation¹⁴, chromium reagents like chromium trioxide-pyridine complex, pyridinium chlorochromate, pyridinium dichromate, or Dess-Martin periodinane¹⁵. Mitter *et al.* reported synthesis of rheinal from 4,5-diacetoxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbonyl chloride [V] *via* Rosenmund reduction¹⁶. Agrawal *et al.* reported the extraction of rheinal from the botanical source *Rheum*

*emodi*¹⁷. Zhang *et al.* reported the oxidation of aloe-emodin to rheinal in presence of TEMPO/trichloro-isocyanuric acid (TCCA) using dichloromethane as solvent. The major disadvantage of this process was hot filtration and requirement of TCCA as the oxidant in stoichiometric amount¹⁸.

In the literature, oxidation of primary benzylic alcohol bearing a protected hydroxyl functionality in the aromatic ring was reported to occur in presence of Cu(II)/TEMPO/ sodium hypochlorite/potassium bromide, and a combination of organo-metallic catalyst obtained from Vanadium (V) and hydrogen peroxide¹⁹⁻²³. However, no reports were available for selective aerial oxidation of primary benzylic alcohol in presence of free hydroxyl functionality in the aromatic ring using Cu(II)/TEMPO and organo-metallic catalyst obtained from Vanadium (V).

Moreover, hydroxy anthraquinones are known to form complexes with various simple metal salts, such as copper, cadmium, nickel, iron, vanadium and cobalt which are called lake complexes^{24,25} and hence simple salts were not employed as catalysts for oxidation.

In the present work, a “green” and eco-friendly method which can be conveniently scaled up to an industrial process for synthesis of rheinal through aerial oxidation in presence of heterogeneous catalysts has been demonstrated.

Results and Discussion

To develop a “green” and eco-friendly process for synthesis of rheinal from aloe-emodin, various catalytic systems were evaluated and it was observed that the following catalytic systems *viz.* VO(acac)₂/DABCO/O₂, Cu(II) Salen H₄ (Ref. 26)/TEMPO/O₂ (Ref. 27), and Laccase²⁸/TEMPO/O₂ gave the desired conversion (**Scheme I**).

For the above catalytic systems *i.e.* VO(acac)₂/DABCO/O₂, Laccase/ TEMPO/O₂ and Cu(II) Salen H₄/TEMPO/ O₂ system conversion of aloe-emodin to rheinal were respectively 100, 48 and 35%. However,

only in the case of Laccase/ TEMPO/O₂ system 34% over-oxidation occurred *i.e.* corresponding carboxylic acid (rhein) was obtained.

In order to increase the rate of oxidation, oxygen pressure was increased from 1 to 3 atm. However, there was no significant improvement in the rate of conversion of aloe-emodin to rheinal.

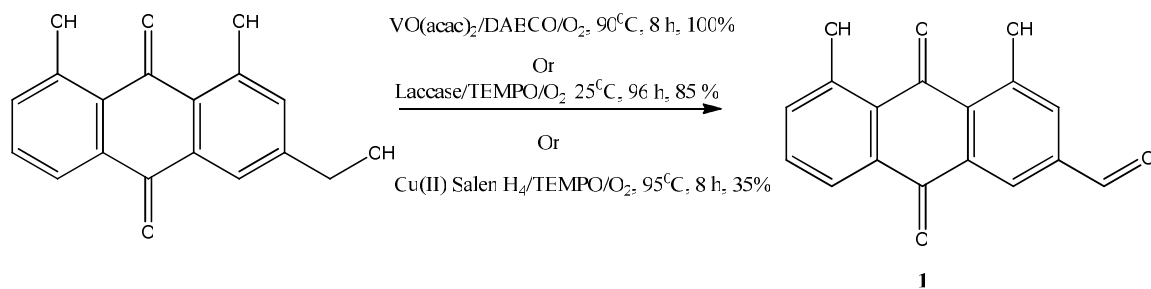
Interestingly, in case of Cu(II) Salen H₄ and VO(acac)₂ oxidation systems, addition of *m*-chloro-perbenzoic acid (*m*CPBA) (10 wt % of substrate) increased the rate of reaction by 2 fold, presumably due to formation of peroxy transition metal complexes^{29,30}.

To understand the role of molecular oxygen, the reaction was carried out using mole equivalent of *m*CPBA in the absence of aerial oxygen. It was observed that under identical conditions, isolated yield for rheinal in presence of VO(acac)₂/DABCO/*m*CPBA (1 equivalent) was only 20% whereas, in case of VO(acac)₂/DABCO/*m*CPBA (10% of substrate)/O₂, it was 82%. Moreover, in case of Cu(II) Salen H₄/TEMPO/ O₂/*m*CPBA, besides an enhanced rate of reaction, the overall conversion of aloe-emodin to rheinal was increased by 2.4 times *i.e.* from 24% (isolated yield) to 58% (isolated yield) without any over oxidized product being formed.

Hence, it can be concluded that a synergistic effect between molecular oxygen and *m*CPBA exists in the present case.

However, when attempts were made to oxidize *p*-hydroxybenzyl alcohol under identical conditions in presence of VO(acac)₂/DABCO/O₂ system, no corresponding oxidation product was formed and the starting material *i.e.* *p*-hydroxybenzyl alcohol was recovered unchanged. Moreover, when the same reaction was attempted under identical conditions by adding 1.1 eq. of *m*CPBA, only 20% of the corresponding aldehyde could be obtained.

In summary, we have developed a novel, efficient and “green” process for oxidation of aloe emodin to rheinal.



Experimental Section

NMR spectra were obtained using 400 MHz Bruker instruments, with DMSO- d_6 as solvent unless otherwise stated. Chemical shifts (δ) are given in ppm relative to tetramethylsilane ($\delta = 0$ ppm). IR spectra were recorded on Perkin-Elmer Spectrum instrument and absorption bands are given in cm^{-1} . Conversion of aloe-emodin to rheinal was determined by HPLC using a Shimadzu LC 2010 system equipped with Inertsil ODS 3V column (4.6×150 mm, $5 \mu\text{m}$), column oven temperature 25°C and UV detector (UV at 254 nm). Mobile phase is buffer: acetonitrile (55:45) with flow rate 1.0 mL min^{-1} , injection volume $20 \mu\text{L}$.

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rheinal) from aloe-emodin through VO(acac) $_2$ /DABCO/O $_2$ oxidation in 1,4-dioxane

A Parr autoclave reactor was charged with 1,4-dioxane (200 mL) at RT. To this were added aloe-emodin (85% purity) (3.0 g), VO(acac) $_2$ (0.15 g) and DABCO (0.2 g) and the reaction mass stirred at RT for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 90°C for 8 hr. Reaction was monitored by TLC (1:1, hexane/ethyl acetate). After complete disappearance of starting material (normally after 8 hr), the reaction mixture was cooled to RT and filtered to remove insoluble materials. The filtrate was poured into 1.0 L of ice-cold water to obtain the product. After filtering and washing with water (200 mL), the collected brown solid was dried at 45°C in vacuum. Isolated yield: 2.0 g (79%); m.p. $202\text{--}205^\circ\text{C}$ (Lit. m.p. 200°C , Ref. 13); Compound 1: ^1H NMR (200 MHz, DMSO- d_6): δ 7.38–8.09 (m, Ar), 10.10 (s, Ar-CHO), 11.98–11.95 (m, Ar-OH); IR: 1699, 1673, 1630, 1265 cm^{-1} .

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rheinal) from aloe-emodin through VO(acac) $_2$ -DABCO/*m*CPBA/O $_2$ oxidation in 1,4-dioxane

A Parr autoclave reactor was charged with 1,4-dioxane (100 mL) at RT. To this were added aloe-emodin (85% purity) (1.5 g), VO(acac) $_2$ (0.07 g), DABCO (0.10 g) and *m*CPBA (0.10 g) and the reaction mass stirred at RT for 15 min. Reactor was

purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 90°C for 5 hr. Reaction was monitored by TLC (1:1, hexane/ethyl acetate). After complete disappearance of starting material (normally after 5 hr), reaction mixture was cooled to RT and filtered to remove insoluble materials. Workup was done in an analogous manner as given previously to obtain compound 1. Isolated yield: 1.05 g (82%).

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rheinal) from aloe-emodin through Cu(II) Salen H $_4$ /TEMPO/O $_2$ oxidation

A Parr autoclave reactor was charged with 1,4-dioxane (100 mL) at RT. To this were added aloe-emodin (85% purity) (1.0 g), Cu(II) Salen H $_4$ (0.15 g) and TEMPO (0.15 g) and the reaction mass stirred at RT for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 95°C for 8 hr and then reaction mixture was cooled to RT and filtered to remove insoluble materials. The filtrate was then poured into 1.0 L of ice-cold water to obtain the product. After filtering and washing with water (200 mL), the collected brown solid was dried at 45°C *in vacuo*. Crude product was further purified by column chromatography using mobile phase *n*-hexane:ethyl acetate (1:1) to obtain the pure aldehyde and unreacted starting material (aloe-emodin) was recovered for later reuse to obtain compound 1. Isolated yield: 0.2 g (24%).

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rheinal) from aloe-emodin through Cu(II) Salen H $_4$ /TEMPO/O $_2$ /*m*CPBA oxidation

A Parr autoclave reactor was charged with 1,4-dioxane (100 mL) at RT. To this were added aloe-emodin (85% purity) (1.5 g), Cu(II) Salen H $_4$ (0.07 g), TEMPO (0.07 g) and *m*CPBA (0.10 g) and the reaction mass stirred at RT for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 95°C for 8 hr. Workup and purification of crude product was done in an analogous manner as given previously to obtain compound 1. Isolated yield: 0.74 (58%).

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rhein-al) from Aloe-emodin through Laccase/TEMPO/O₂ oxidation

A Parr autoclave reactor was charged with 10 mmol sodium acetate buffer of pH 5.0 (100 mL) to which were added aloe-emodin (85% purity) (3.0 g), Laccase from *T. versicolor* (0.5 g) and TEMPO (0.9 g) and the reaction mass stirred at RT for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 25°C for 96 hr. Reaction was monitored by HPLC to give 34% yield of the corresponding carboxylic acid and 48% yield of the corresponding aldehyde and 15% unreacted aloe-emodin.

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rhein-al) from aloe-emodin through Cu(OAc)₂/TEMPO/O₂ oxidation in 1,4-dioxane

A par autoclave reactor was charged with 1,4-dioxane (200 mL) at RT. To this were added aloe-emodin (85% purity) (2.0 g), Cu(OAc)₂ (0.07 g) and TEMPO (0.56 g) and the reaction mass stirred at RT for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in the reactor. The resulting mixture was stirred at 90°C for 8 hr. Reaction was monitored by TLC (1:1, hexane/ethylacetate) wherein no oxidized product was observed.

Process for preparation of 4,5-dihydro-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rhein-al) from aloe-emodin through VO(acac)₂-DABCO/*m*CPBA/ oxidation in 1,4-dioxane

A Parr autoclave reactor was charged with 1,4-dioxane (100 mL) at RT. To this were added aloe-emodin (85% purity) (1.5 g), VO (acac)₂ (0.07 g), DABCO (0.10 g) and *m*CPBA (1.0 g). The resulting mixture was stirred at 90°C for 8 hr. Thereafter, the reaction mixture was cooled to RT and filtered to remove insoluble materials. Workup was carried out in an analogous manner as described previously. Crude product was further purified by column chromatography using mobile phase hexane:ethyl acetate (1:1) to obtain the pure aldehyde and

unreacted starting material (aloe-emodin). Isolated yield: 0.3 g (20%).

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